

Akash Karthik

San Jose, CA - akashkarthik473@gmail.com - linkedin.com/in/akash-karthik473 - github.com/akashkarthik473 - 9254069507

EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: C, C++, Python, Assembly, Java

Embedded: Real-time firmware, microcontroller I/O, TTC 60 VCU, STM32H7, PX4, on-vehicle bring-up and debug

Protocols & Buses: CAN, SPI, DBC signal definitions, CAN diagnostics, bus capture and trace analysis, firmware flashing over CAN

Firmware & Controls: Fixed-point PI/PID, deterministic state machines, anti-windup, slew-rate limiting, sensor fusion, control-loop tuning

Tools: TASKING, TTC Downloader, BusMaster, MoTeC i2 Pro, GDB, Git, CSV telemetry

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Leading an in-house VCU program replacing a vendor controller with custom STM32H755 hardware, running a 10 ms deterministic control task under a watchdog
- Built its CAN driver across both controller buses and specified the analog front-end changes needed for vehicle bring-up

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated embedded flight-control firmware in C on PX4, iterating control gains against logged flight telemetry
- Tuned PID flight-control loops on a PX4 flight controller to reduce vertical and horizontal position drift in autonomous flight
- Integrated and tuned visual-inertial odometry for onboard state estimation, improving position-hold accuracy

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Developed embedded power-limiting firmware with a PI controller on the VCU, maximizing motor output while enforcing FSAE's 80 kW limit
- Implemented a torque-to-power controller with fixed-point PI control, saturation, anti-windup, and slew limits for deterministic VCU execution
- Built CAN diagnostics firmware for live power and torque frames, status bits, and error counters for real-time BusMaster monitoring

Software Intern

Aug 2024 - Jun 2025

Spartan Racing, Formula SAE | San Jose, CA

- Built deterministic entry and exit state logic for power limiting from measured power, requested torque, and threshold flags
 - Tuned the PI torque controller used for power limiting to within **2%** error, smoothing torque transitions to preserve drivability
 - **Results:** 1st in endurance at MIS EV 2025
-

PROJECT EXPERIENCE

SR-Wireless-CAN — Wireless Vehicle Telemetry Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, python-can, FastAPI, WebSockets, SQLite, Docker

- Proposed and implemented wireless firmware flashing for the TTC 60 VCU, decoding the flashing sequence from CAN trace captures and reimplementing it in Python over the bus
- Developed wireless CAN capture and diagnostics tooling with DBC-driven signal decoding, replacing tethered bench monitoring during vehicle testing

Formula SAE EV Dashboard

Feb 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

C, STM32H7, CAN, SPI

- Built a real-time CAN telemetry display for HV voltage, throttle, cell temperature, and power budget on an LCD
- Designed an energy bar and LED alert system giving the driver lap-by-lap feedback on energy budget